More Bode Sketching and Start of Bode ID

EGR 345

GVSU

Outline

- groups 1, 2, 4, and 5 present Bode sketching examples
- start Bode system ID

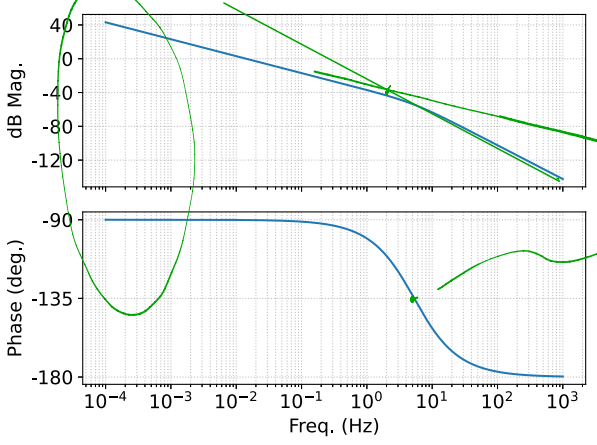
Bode System ID

- what does Bode system ID mean?

→ finding a TF by reading the Bode plot
→ form (poles and zeros)
→ find #'s for coeffs.

Bode ID Example 1

- what is the TF for this Bode plot?



Bode ID Algorithm

- identify the low frequency behavior
  - is there a pure int., a pure diff. or neither?
    - first or second order pure int. or diff.?
- identify each frequency where there is a change in phase and slope
  - phase and slope should always change together
  - sometimes one will be more obvious than the other
  - the phase is sometimes more helpful
- when the slope and phase change, determine if it is a pole or zero
  - is it first or second order?
- continue until you get to above the highest slope and phase change
- multiply the terms together to get the full TF

* do the phase and slope increment or decrement?*

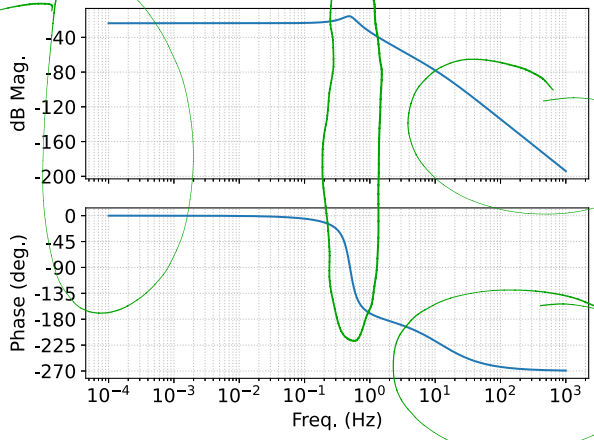
write the TF terms for each pole or zero

Example 1 Python Verification

- grab csv file: csv_files from class 18 github
- grab starter notebook: bode_system_ID_verification.ipynb

Bode ID Example 2

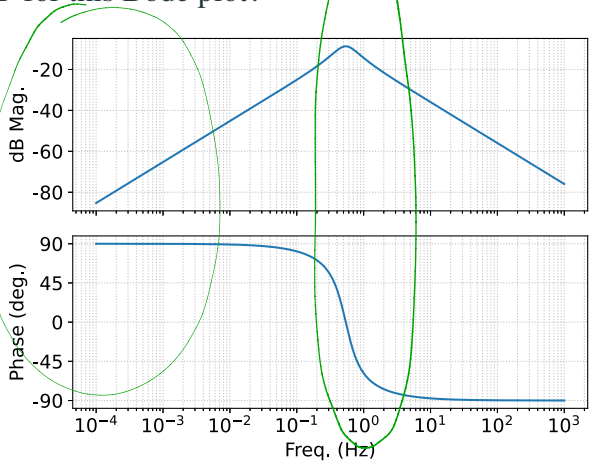
- what is the TF for this Bode plot?



Verification for Example 2

Bode ID Example 3

- what is the TF for this Bode plot?



Verification for Example 3

Random Bode ID Example(s)

- Bode ID use case for `rbg`

→ create random TF, but don't look at it

→ plot Bode